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Improvement in the oxidation resistance of Nb–Ti–Si–Cr–Al–Hf alloys containing alloyed Ge and B

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ABSTRACT

The oxidation behaviour for Nb–Si based alloys was performed in air at 1200 °C and 1250 °C. The weight gains of Nb–24Ti–15Si–13Cr–2Al–2Hf (at.%) alloy and Nb–24Ti–15Si–13Cr–2Al–2Hf–4B–5Ge (at.%) alloy after oxidation at 1250 °C for 100 h were 156.7 mg/cm^2 and 32.5 mg/cm^2, respectively. The oxide scale of the Nb–24Ti–15Si–13Cr–2Al–2Hf (at.%) alloy was porous and loose, while that of the Nb–24Ti–15Si–13Cr–2Al–2Hf–4B–5Ge (at.%) alloy was dense and protective. The presence of alloyed B and Ge improved the protective SiO_2 phase grown during high temperature oxidation, leading to the formation of a continuous and dense oxide scale.

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1. Introduction

The Nb–Si based alloys have been widely studied recently due to their attractive properties such as low density, high melting point and high temperature strength [1–3]. However, the widespread use of Nb–Si based alloys in aircraft, missiles, and certain reactors is limited because their oxidation resistance is still inadequate for structural applications [1–4].

In order to achieve a balance of properties for structural use such as high temperature strength and good oxidation resistance, a multi-component Nb–Si–Ti–Hf–Cr–Al system has been developed by General Electric Co. [2,3]. In this system, the phase constitutions are silicides, Cr_2Nb and Nb solid solution (Nb_{SS}) (the elements mainly solute in Nb are Ti and Cr). This system is of interest because the low temperature fracture toughness can be improved by Nb_{SS} and the high temperature strength and good oxidation resistance can be anticipated from the silicides and Laves phase [5–9].

One of the difficulties encountered in the oxidation resistance of Nb–Si based alloys is the formation of non-protective oxides (Nb_2O_5, $CrNbO_4$, $Ti_2Nb_{10}O_{29}$ and $Nb_2O_5{\cdot}TiO_2$) at high temperature [5,9,10]. Therefore, it is necessary to promote the formation of a protective oxide layer on the surface of Nb–Si based alloys. Extensive studies have shown that the oxidation resistance of Nb–Si based alloys can be improved significantly with the addition of Cr, Si, Al, Ti and Hf [2–4,9–12]. Alloying with B can be beneficial to the oxidation resistance [13]. Boron tends to interstitially soluble to form $Nb_5Si_3B_x$ with $D8_8$ structure (Mn_5Si_3-type, hexagonal). At the same time, when B replaces with Si, the $Nb_5(Si, B)_3$ phase is formed. The $Nb_5(Si, B)_3$ phase, i.e., so called "T_2 phase" has a body-centered tetragonal Cr_5B_3-type structure [14]. As suggested by Behrani et al., boron containing phases (T_2 and $D8_8$) show better oxidation resistance than Nb_5Si_3 [13]. Germanium addition, as a partial replacement of Si, also improves the oxidation behaviour of Nb–Si based alloy [12]. The Ge-doped silicide coating protects the refractory metals from significant oxidation and pesting by the formation of a Ge-doped silica film [15,16].

Investigation of Nb–Si based alloys without coatings is crucial to determine how they will respond in case of a failure of the coating. And understanding the oxidation kinetics of Nb–Si based alloys is of crucial importance for improving the oxidation resistance. In present study, the microstructure and oxidation behaviour of Nb–Si based alloy with compositions Nb–24Ti–15Si–13Cr–2Al–2Hf (at.%) and Nb–24Ti–15Si–13Cr–2Al–2Hf–4B–5Ge (at.%) were investigated. The results provide possibilities to develop a structural material with improved oxidation resistance for high temperature applications.

2. Experimental procedures

2.1. Specimen preparation

Two alloys with compositions of Nb–24Ti–15Si–13Cr–2Al–2Hf (at.%) (Alloy 1) and Nb–24Ti–15Si–13Cr–2Al–2Hf–4B–5Ge (at.%) (Alloy 2) were prepared by arc melting, a mixture of high purity niobium, titanium, chromium, aluminium, silicon, hafnium, boron

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and germanium in arc furnace under an argon atmosphere. To ensure composition homogeneity and element partition, the ingots were inverted and remelted four times. Oxidation specimens with a size of $8 \times 8 \times 5$ mm^3 were cut from the ingots by electron discharge machine and all surfaces were mechanically ground with 800-grit SiC paper and ultrasonically cleaned.

2.2. Oxidation testing

The oxidation tests were performed in an open-ended tube furnace at 1200 °C and 1250 °C. Each specimen was placed in a separate alumina crucible during the test. The specimens were removed from the furnace at the intervals of 10, 20, 40, 60, 80 and 100 h and weighed together with the crucible using a precision analytical balance (Model CPA225D, Germany) with an accuracy of 0.00001 g.

2.3. Mechanical properties testing

Compression specimens with a size of $4 \times 4 \times 6$ mm^3 were cut from the ingots by electron discharge machine and all surfaces were mechanically ground with 600-grit SiC paper prior to the experiments. Room temperature compression tests were conducted at room temperature in air at an initial strain rate of 0.09 mm/min on a SANS 5104 testing machine. High temperature compression tests were conducted at 1200 °C in argon at an initial strain rate of 0.01 mm/min on a Gleeble 1500 testing machine.

2.4. Analysing methods

X-ray diffraction (XRD) (Model O/MAX-2200, Japan) pattern of the oxidation products was investigated to identify the main phases. The surface morphologies of oxidised specimens were analysed in scanning electron microscopes (SEM) (Model CS-3400, Japan). The cross-sectional microstructure of oxidised specimens were analysed in electron probe micro-analyzer (EPMA, Model JXA-8230, Japan) equipped with wave-dispersive spectroscopy (WDS) using backscattered scanning electron (BSE) and X-ray mapping modes. Nb L_α, Si K_α, Cr K_α, Ti K_α, Al K_α, Hf L_α, B K_α, Ge K_α, O K_α were analysed with PETJ, TAP, LIFH, PETJ, LIFH, PETJ, TAP, LIFH, LDE2H, LIFH and LDE2H crystals, respectively. The ZAF corrected EPMA was calibrated by pure standards supplied by the manufacture for different operating conditions and probe sizes. Before making observations, the cross sections were cut and ground with 1500-grit SiC paper, and then polished on a tightly woven cloth with 1 μm diamond paste.

3. Results

3.1. Microstructural characterization of as-cast alloys

Fig. 1a and b demonstrate the microstructure of Alloy 1 and Alloy 2 respectively. And the XRD patterns of two alloys are shown in Fig. 1c, where the T_2 phase and $D8_8$ phase are identified according to the XRD peaks of the α-Nb_5Si_3 phase and γ-Nb_5Si_3 phase respectively [14]. All phases have been confirmed through the use of XRD and WDS analysis. Alloy 1 consists of dark grey α-Nb_5Si_3, white Nb_{SS} and black Cr_2Nb, while Alloy 2 contains light grey $Nb_5(Si, Ge)B_2$ (T_2 phase), dark grey $Nb_5(Si, Ge)_3$ ($D8_8$ phase), white Nb_{SS} and black Cr_2Nb. The EPMA analysis results of as-cast alloys are given in Table 1. As shown in Table 1, (Nb + Ti)/(Si + Ge + B) atomic ratio is close to 5:3 in the T_2 phase, while (Nb + Ti)/(Si + Ge) atomic ratio is close to 5:3 in the $D8_8$ phase. This result suggests that Ge replaces with Si in both T_2 and $D8_8$ phase, and B replaces with Si in T_2 phase but interstitially solutes in $D8_8$ phase.

3.2. Oxidation kinetics

Fig. 2a shows the weight gain per unit area as a function of the exposure time at 1200 °C and 1250 °C for two alloys. Alloy 1 displays a linear plot with the weight gains of 118.7 mg/cm^2 and 156.7 mg/cm^2 after oxidation at 1200 °C and 1250 °C respectively. Alloy 2 shows better oxidation resistance with the weight gains of 26.7 mg/cm^2 and 32.5 mg/cm^2 after oxidation at 1200 °C and 1250 °C respectively. Fig. 2b presents the weight gains of Alloy 2 as a function of the square root of exposure time after oxidation

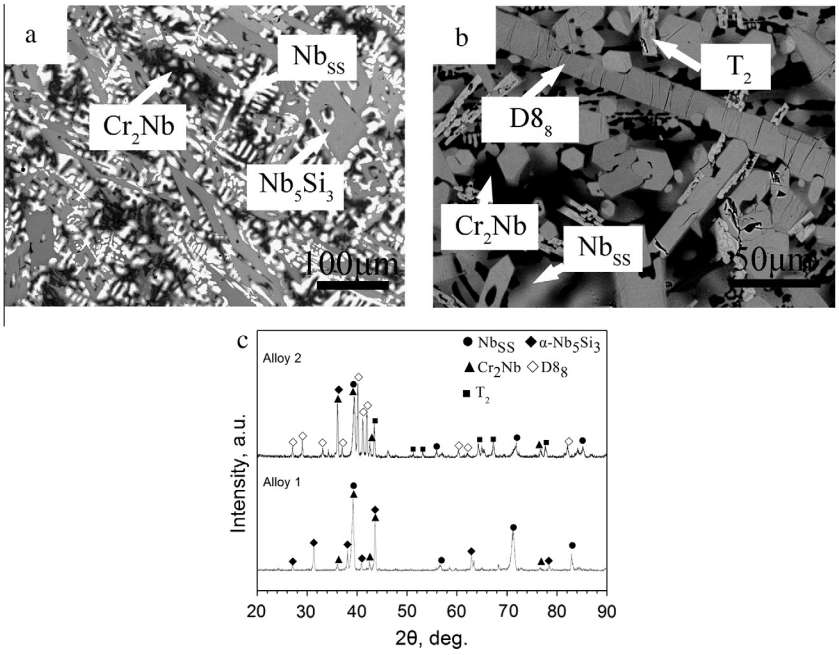


Fig. 1. (a) Microstructure of Nb–24Ti–15Si–13Cr–2Al–2Hf (at.%) alloy; (b) microstructure of Nb–24Ti–15Si–13Cr–2Al–2Hf–4B–5Ge (at.%) alloy; (c) XRD patterns of the specimens.

Table 1
Phase compositions of the as-cast alloys (at.%).

	Si	Al	Nb	Cr	Hf	Ti	Ge	B
Alloy 1								
Nb_{SS}	1.36	3.06	51.12	14.74	0.88	28.84	–	–
Nb_5Si_3	38.27	0.83	42.83	0.45	1.55	16.07	–	–
Cr_2Nb	5.39	1.04	19.84	52.58	2.54	18.61	–	–
Alloy 2								
Nb_{SS}	0.80	3.77	44.36	16.71	0.63	33.02	0.73	0
Cr_2Nb	7.25	1.18	21.20	51.55	1.96	16.18	0.88	0.04
T_2	11.95	0.01	44.27	3.75	0.58	11.54	2.05	25.87
$D8_8$	27.73	1.81	34.60	1.53	2.52	21.79	7.04	3.59

at 1200 °C and 1250 °C, which are nearly linear, indicating that Alloy 2 presents the parabolic plot. The parabolic rate constants are calculated according to the equation:

$$\left(\frac{\Delta m}{A}\right)^2 = k_p t \quad (1)$$

where Δm is the weight change of the specimen, A is the surface area and t is the exposure time. The parabolic rate constants ($g^2\,cm^{-4}\,s^{-1}$) calculated for Alloy 2 after oxidation at 1200 °C and 1250 °C for 100 h are equal to 1.90×10^{-9} and 2.80×10^{-9} respectively.

3.3. Structures and morphologies of oxide scale

3.3.1. Scale structures

Fig. 3 shows XRD patterns of oxidised products and it appears to reveal considerable amounts of information on the oxidation characteristics. It is clear that Nb_2O_5, TiO_2, $CrNbO_4$ and $TiNb_2O_7$ phases are formed on Alloy 1 and Alloy 2 after oxidation. The intensity ratio of the most intense reflection for $CrNbO_4$ and Nb_2O_5 (I_{CrNbO4}/I_{Nb2O5}) is used to perform a comparison of the relative amounts of $CrNbO_4$ and Nb_2O_5 in the oxidised products. It could be found that the intensity ratio (I_{CrNbO4}/I_{Nb2O5}) of Alloy 2 is higher than that of Alloy 1, indicating that the predominant oxide formed on Alloy 2 is $CrNbO_4$, which is expected to be a protective oxide to enhance the oxidation resistance of the Nb–Si based alloy.

3.3.2. Surface morphologies

Fig. 4 shows SEM images of the surface morphologies of Alloy 1 and Alloy 2 after oxidation at 1200 °C and 1250 °C for 100 h. It is noted that the surfaces formed on Alloy 1 are rough and porous (as shown in Fig. 4a and b), which is due to the serious spallation of oxidised products during oxidation. For Alloy 2, the scales are basically intact. As shown in Fig. 4c, the surface is composed of granular oxide structure (~4 μm) with some rod-like phases distributed on it. And the surface is composed of coarse granular oxide structure (~10 μm) after oxidation at 1250 °C (as shown in Fig. 4d).

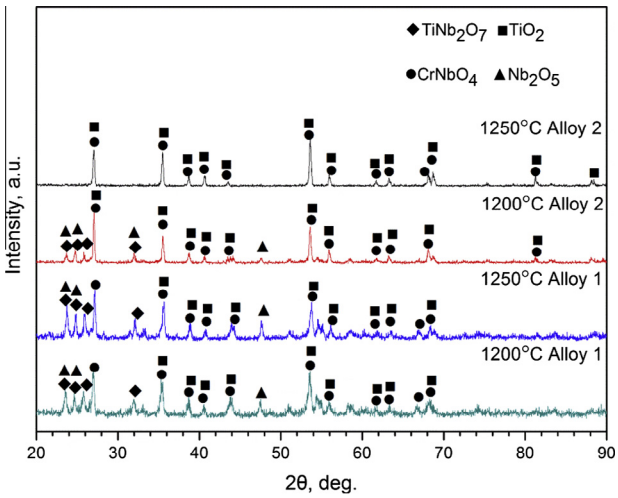


Fig. 3. XRD patterns of oxides obtained after oxidation at 1200 °C and 1250 °C for 100 h.

3.3.3. Cross-sectional microstructure

The SEM images of cross-sectional microstructure of Alloy 1 and Alloy 2 oxidised at 1200 °C and 1250 °C are presented in Fig. 5. The oxidised products are identified based on the results from the XRD analysis, the WDS analysis and BSE contrast. The oxide scale formed on the surface of Alloy 1 is porous and loose with cracks. Besides, the oxide scale formed at 1250 °C contains more cracks compared to that formed at 1200 °C, which is due to the faster oxidation rate at a higher oxidation temperature. At higher temperature, the oxide scale formed on the niobium alloy was non-adherent and easily flaked off [17]. The WDS analysis in Fig. 5a reveals that the dark-region has a composition of 21.84Nb–4.51Ti–1.01Cr–0.48Hf–0.31Al–71.16O (at.%), while the light-grey regions has a composition of 9.98Nb–12.16Ti–8.47Cr–0.53Hf–1.53Al–67.16O (at.%). Combined with Fig. 3, it indicates that they are Nb_2O_5 and the mixture of TiO_2, $CrNbO_4$ and $TiNb_2O_7$, respectively.

But for Alloy 2, a uniform and protective oxide scale is formed on the surface. The oxide scale remains adhered to the substrate and no peeling occurs during the cooling, which suggests that the oxide scale has good adherence to the substrate. Besides, the

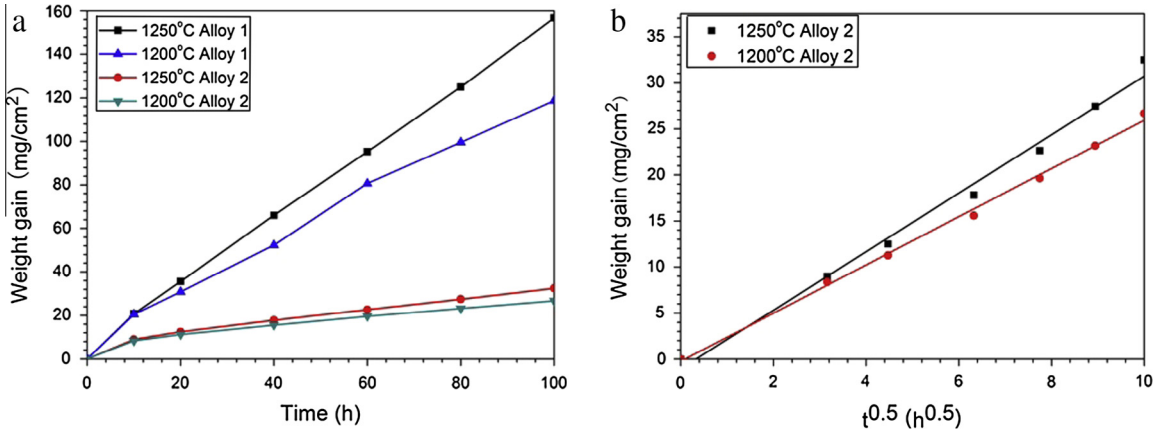


Fig. 2. (a) Weight gain versus time for Alloy 1 and Alloy 2 oxidised in air; and (b) representation of the weight gain versus the square root of time for Alloy 2 oxidised in air.

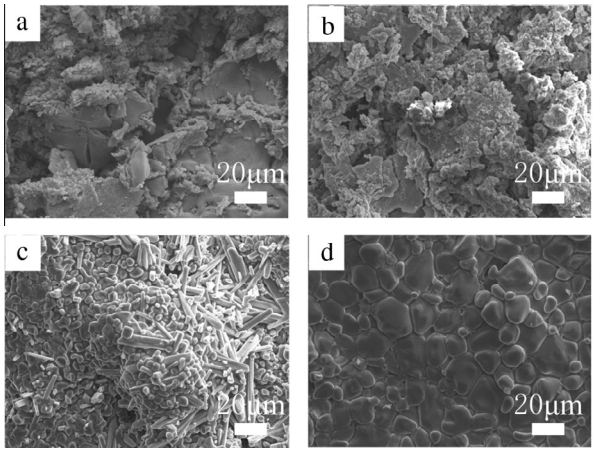


Fig. 4. Surface morphologies of Nb–24Ti–15Si–13Cr–2Al–2Hf alloy (at.%) after oxidation for 100 h at: (a) 1200 °C; (b) 1250 °C; surface morphologies of Nb–24Ti–15Si–13Cr–2Al–2Hf–4B–5Ge (at.%) alloy after oxidation for 100 h at: (c) 1200 °C; (d) 1250 °C.

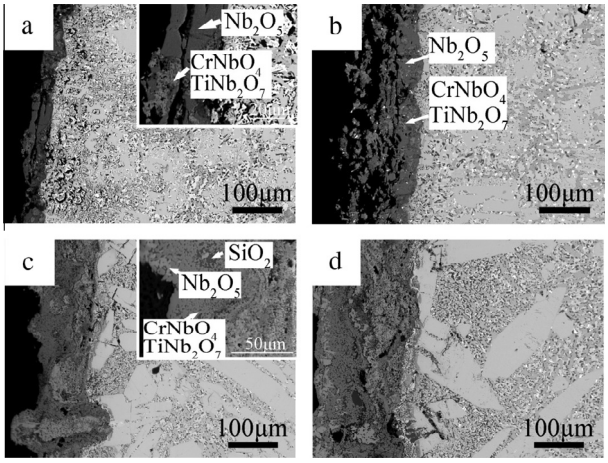


Fig. 5. Cross-sectional microstructure of Nb–24Ti–15Si–13Cr–2Al–2Hf (at.%) alloy after oxidation for 100 h at: (a) 1200 °C and (b) 1250 °C; cross-sectional microstructure of Nb–24Ti–15Si–13Cr–2Al–2Hf–4B–5Ge (at.%) after oxidation for 100 h at: (c) 1200 °C and (d) 1250 °C.

thicknesses of oxide scale are 100 and 200 μm for Alloy 2 after oxidation at 1200 °C and 1250 °C respectively. The oxide scale of Alloy 2 could be divided into two layers. The outer layer is dense and continuous while the inner layer is porous and loose. The WDS analysis in Fig. 5c reveals that the dark phase distributed in the space of granular oxide structure with dark grey contrast has a composition of 5.73Nb–4.30Ti–3.99Cr–0.47Hf–2.20Al–15.11Si–2.04Ge–66.16O (at.%), while the granular oxide structure with light grey contrast has a composition of 10.18Nb–14.55Ti–7.90Cr–0.70Hf–1.40Al–0.33Si–0.04Ge–64.90O (at.%). Combined with Fig. 3, it indicates that they are SiO_2 and the mixture of $CrNbO_4$ and $TiNb_2O_7$, respectively. In this alloy, SiO_2 is observed in the image of cross-sectional microstructure (Fig. 5c) but the XRD pattern for the oxidised products shows no sharp peaks of SiO_2, which indicates that SiO_2 has the structure of amorphous glass. WDS analysis indicates that in the inner layer, the dark-grey phase (7.95Nb–18.77Ti–9.21Cr–0.38Hf–0.13Al–0.54Si–0.09Ge–62.93O (at.%)) could be the mixture of $CrNbO_4$ and TiO_2 while the light-grey phase (23.86Nb–10.44Ti–0.11Cr–0.42Hf–0.80Al–0.33Si–0.22Ge–63.82O (at.%)) could be $TiNb_2O_7$.

Fig. 6 shows the elemental X-ray mapping of Alloy 2 oxidised at 1250 °C for 100 h obtained using WDS attached with EPMA. The results clearly show the presence of oxygen, niobium, silicon,

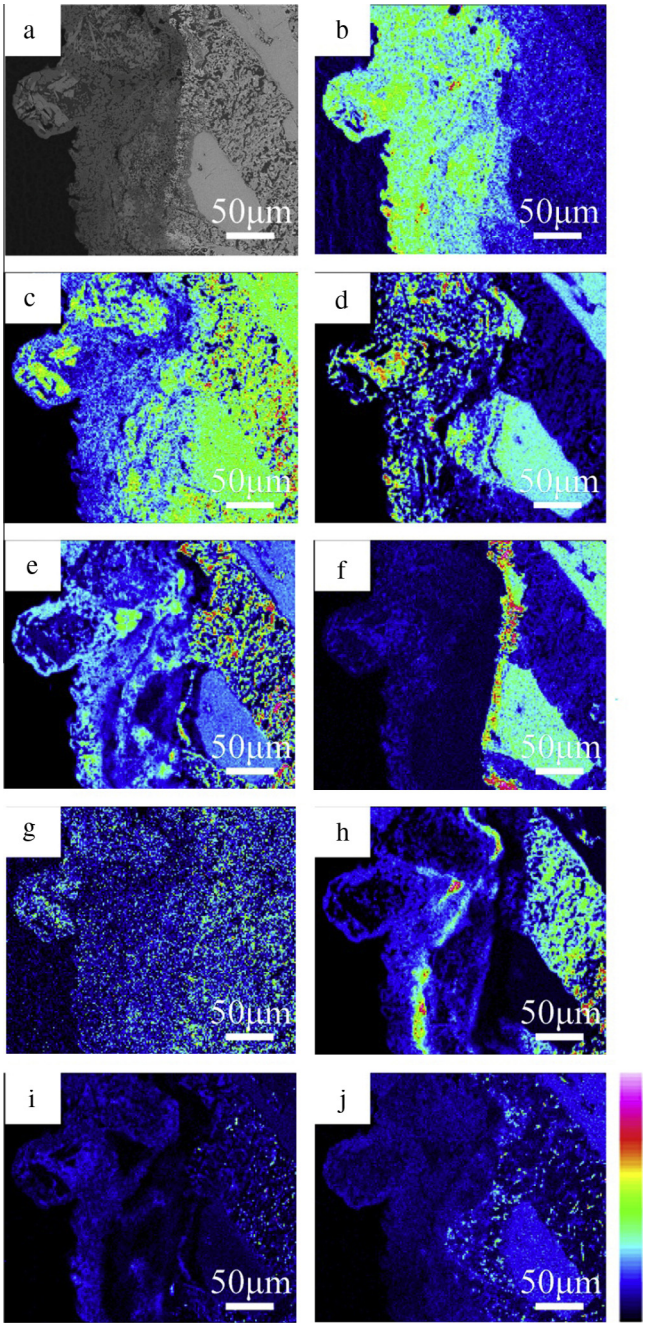


Fig. 6. Elemental X-ray mapping of Alloy 2 after oxidation at 1250 °C for 100 h obtained using WDS attached with EPMA (a) cross-sectional microstructure; (b) O mapping; (c) Nb mapping; (d) Si mapping; (e) Ti mapping; (f) Ge mapping; (g) B mapping; (h) Cr mapping; (i) Al mapping; (j) Hf mapping.

titanium, germanium, boron, chromium, aluminium and hafnium on the oxide layer. The colour scale is given in the bottom right corner: red represents the maximum concentration, yellow and green represent moderate concentrations, while blue represents a low concentration. As shown in Fig. 6f, the intermediate oxide–metal interface is found to be relatively rich in Ge. Besides, Ge intends to solve in the outer layer of the oxide scale. Although it is difficult to obtain an accurate boron content in oxide layer by quantitative analysis due to the low concentration of boron, the EPMA mapping shows the presence of boron on the oxide layer (as shown in Fig. 6g). In addition, the chromium (Fig. 6h) and aluminium (Fig. 6i) depleted regions are found in the oxide–metal interface, suggesting that the oxidation occurs not only by inward diffusion of O but also by outward diffusion of Cr and Al.

3.3.4. Internal oxidation zone

Fig. 7 shows the microstructure of the internal oxidation zone of two alloys after oxidation at 1250 °C for 100 h. The diffusion of oxygen through the Nb_{SS} leads to the formation of a thick zone affected by internal oxidation. Alloy 1 is completely affected by the internal oxidation at 1200 °C and 1250 °C. For Alloy 2, the internal oxidation zone is approximately 1500 μm after oxidation at 1200 °C for 100 h. And it is completely affected by the internal oxidation at 1250 °C. In the internal oxidation zone, TiO_2 with black contrast and HfO_2 with white contrast are observed at the Nb_{SS} and the interface of Nb_{SS}/Nb_5Si_3. Oxygen would react with Ti and Hf due to their high affinities with oxygen. Mathieu et al. reported that the outer part of silicide was fully oxidised in the internal oxidation zone, forming a mixture of oxides consisting of TiO_2 and HfO_2 [18]. But in this study, the silicides remain unaffected by oxygen penetration. In addition, the internal oxidation zone of Alloy 1 is thicker than that of Alloy 2 after oxidation at 1200 °C, suggesting that the inward diffusion of oxygen is faster in Alloy 1 than that in Alloy 2. Besides, the thickness of internal oxidation zone increases significantly with the increase of temperature, which may be due to the faster inward diffusion of oxygen through the Nb_{SS} at higher temperature.

3.4. Mechanical properties

Compression tests were conducted to investigate the mechanical properties of Alloy 1 and Alloy 2 at room temperature in air as well as at 1200 °C in argon. As shown in Table 2, the room temperature yield strengths of Alloy 1 and Alloy 2 are 2185.3 MPa and 1554.0 MPa respectively, while the yield strengths of Alloy 1 and Alloy 2 are 206.4 MPa and 306.6 MPa respectively. Sekido et al. [19] has investigated the room temperature fracture toughness and high temperature strength of Nb–Si based alloys. The results showed that as the volume fractions of Nb_{SS} increased, the fracture toughness increased but the high temperature strength decreased. In this study, the volume fraction of Nb_{SS} in Alloy 1 is higher than that in Alloy 2. Therefore, Alloy 1 exhibits higher room temperature yield strength but lower high temperature strength compared to Alloy 2. As designed by Jackson et al. [20], a turbine component may comprise a core and a surface layer thereon. The core of the turbine component provided enhanced creep resistance and toughness characteristics while the surface layer provided enhanced oxidation. As Alloy 2 shows good oxidation resistance and acceptable mechanical properties, it can be used as the promising surface layer material of the turbine.

4. Discussion

4.1. Oxidation behaviour of Nb–24Ti–15Si–13Cr–2Al–2Hf (at.%) alloy (Alloy 1)

As described in Section 3.3.3, the oxide scale formed on Alloy 1 consists of Nb_2O_5 and the mixture oxides ($TiNb_2O_7$ and $CrNbO_4$), which is porous with many cracks formed on the Nb_2O_5. The oxidation reactions of the mixture oxide may be written as follows [21]:

$$Nb_2O_5 + TiO_2 \rightarrow TiNb_2O_7 \quad (2)$$

$$Nb_2O_5 + Cr_2O_3 \rightarrow CrNbO_4 \quad (3)$$

And the worse oxidation resistance of this alloy may be due to the larger amount of Nb_2O_5. The formation of Nb_2O_5 involves significant volume expansion of 169% [22]. The voluminous oxides can create high compressive stresses that crack the oxide layer and expose a new surface area [23], resulting in the spallation of oxide scale. Besides, no SiO_2 appears in the oxide scale. In the study of Varma [6], SiO_2 appeared in the oxide scale of Nb–20Cr–(10-42Si) (at.%) alloys when the concentration of Si was higher than 30 at.%. The reason for the no presence of SiO_2 may be due to the lower percentage of Si in Alloy 1 (15 at.%).

Table 2
Compressive strengths of Alloy 1 and Alloy 2.

Room-temperature yield strength in air (MPa)	Yield strength at 1200 °C in argon (MPa)
Alloy 1	
2221.1	212.3
2149.4	200.4
Average	
2185.3	206.4
Alloy 2	
1435.0	236.3
1673.0	376.8
Average	
1554.0	306.6

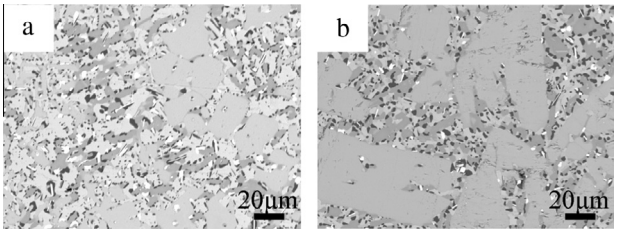


Fig. 7. Microstructure of the internal oxidation zone after oxidation at 1250 °C for 100 h: (a) Alloy 1; (b) Alloy 2.

4.2. Oxidation behaviour of Nb–24Ti–15Si–13Cr–2Al–2Hf–4B–5Ge (at.%) alloy (Alloy 2)

It should be noticed that SiO_2 appeared in the outer layer of Alloy 2 besides the mixture oxides of $CrNbO_4$ and $TiNb_2O_7$. The presence of SiO_2 could self-cure the oxide scale at high temperature, generating a continuous and dense outer layer. As reported by Bewlay et al. [3], Ge and B could partially substitute for Si in Nb–Si based alloy. As Alloy 2 is alloyed with B and Ge, the percentage of Si in Alloy 2 could be considered higher than that in Alloy 1, leading to the formation of SiO_2 after oxidation. Many investigations have been done in order to form a continuous SiO_2 layer in the oxide scale of Nb–Si based alloy; however, it is difficult to form such a continuous layer. As reported by Murakami et al. [24], the oxide layer on the Nb–34Si–10B (at.%) alloy fabricated by spark plasma sintering consisted of Nb_2O_5 and crystalline SiO_2 with plenty of pores, but no continuous SiO_2 layer existed. It is interesting to note that the continuous SiO_2 phase is obtained in Alloy 2. As shown in Fig. 6, germanium and boron present in the oxide layer. Furthermore, a germanium rich area is observed in the oxide–metal interface. It is reasonable to believe that germanium and boron dissolve in the oxide layer, which could improve the protective SiO_2 phase grown during high temperature oxidation by forming a more fluid glass and healing cracks and pores at a lower temperature [25]. Thus, the present of alloyed boron and germanium facilitates a faster coverage of a continuous SiO_2 phase on the oxide scale, leading to a more limited oxygen diffusion rate through the scale and consequently yielding a slower growth of oxide scale. Additionally, B_2O_3 and GeO_2 dissolved in silica glass dramatically increase the coefficients of thermal expansion (CTE) of scale. This increase produces a much better match in CTE with the Nb–Si based alloy, minimizing spallations and cracks of the oxide scale upon thermal cycling [26].

However, the inner layer of Alloy 2 consists of $CrNbO_4$, $TiNb_2O_7$ and TiO_2 but no evidence of SiO_2. The silicon mapping (Fig. 6d) can be used as the evidence that the formation of SiO_2 in the outer layer is creating a concentration gradient at the interface. Thus, the silica is unable to form due to inadequate silicon in the inner layer. Therefore, the inner part of oxide layer cannot be self-cured, resulting in a more porous and loose scale.

In summary, the present of alloyed boron and germanium improves the protective SiO_2 phase grown during high temperature oxidation by forming a more fluid glass and healing cracks and pores at a lower temperature. The formation of SiO_2 decelerates the diffusion of oxygen into the metal and decreases the spallation of the oxide scale.

5. Conclusions

1. The microstructure of Nb–24Ti–15Si–13Cr–2Al–2Hf (at.%) alloy consisted of Nb_5Si_3, Nb_{SS} and Cr_2Nb, while the microstructure of Nb–24Ti–15Si–13Cr–2Al–2Hf–4B–5Ge (at.%) alloy consisted of Nb_{SS}, $Nb_5(Si, Ge)_3$ ($D8_8$ phase), $Nb_5(Si, Ge)B_2$ (T_2 phase) and Cr_2Nb.
2. The weight gain of Nb–24Ti–15Si–13Cr–2Al–2Hf–4B–5Ge (at.%) alloy was 32.5 mg/cm^2 after oxidation at 1250 °C for 100 h, which was a quarter of the value of that for Nb–24Ti–15Si–13Cr–2Al–2Hf (at.%) alloy. The presence of alloyed B and Ge improved the protective SiO_2 phase grown during high temperature oxidation, leading to the formation of a continuous and dense oxide scale on the surface of the alloy.

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